

Fpt University
TRƯỜNG ĐẠI HỌC FPT

MINISTRY OF EDUCATION AND TRAINING

FPT UNIVERSITY

Capstone Project Document

[Project name]

<Group Name>	
Group Members	<Member name><RollNo><Student code > <Member name><RollNo>< Student code > <Member name><RollNo>< Student code > <Member name><RollNo>< Student code > <Member name><RollNo>< Student code >
Supervisor	

Ext Supervisor	
Capstone Project code	

- Hanoi, <month>/<year> -

Table of Contents

Acknowledgement

Definition and Acronyms

I. Project Introduction

- 1. Overview
- 2. Product Background
- 3. Existing Systems
- 4. Business Opportunity
- 5. Software Product Vision
- 6. Project Scope & Limitations

II. Project Management Plan

- 1. Overview
- 2. Management Approach
- 3. Master Schedule
- 4. Project Organization
- 5. Project Communication
- 6. Configuration Management

III. Software Requirement Specification

- 1. Overall Description
- 2. User Requirements

3. Functional Requirements

4. Non-Functional Requirements

5. Other Requirements

IV. Software Design Description

1. Overall Description

2. System Architecture Design

3. System Detailed Design

4. Class Specification

5. Data & Database Design

V. Software Testing Documentation

1. Overall Description

2. Test Plan

3. Test Cases

4. Test Reports

VI. Release Package & User Guides

1. Deliverable Package

2. Installation Guides

3. User Manual

VII. Appendix

1. Glossary [Optional]

2. References [Optional]

3. Others [Optional]

Acknowledgement

[Fill team's acknowledgement here…]

Definition and Acronyms

[Fill all the definitions, acronyms,… used within the document] in the table format as below]

Acronym	Definition
PWM	Psychology website
AWS	Amazon Web Services
BA	Business Analysis
BR	Business Rule
ERD	Entity Relationship Diagram
GUI	Graphical User Interface
PM	Project Manager
SDD	Software Design Description
SPMP	Software Project Management Plan
SRS	Software Requirement Specification
UAT	User Acceptance Test
UC	Use Case
API	Application Program Interface

I. Project Introduction

1. Overview

1.1 Project Information

Project name: AI-powered Personalized Learning Path Platform for Competency Assessment Exam Preparation
Project code: FA26SE090
Group name: FA26SE090
Software type: Web, Mobile App

1.2 Project Team

Full Name	Role	Email	Mobile
Nguyễn Trọng Tài	Lecturer	taint@fpt.edu.vn	
Phạm Tấn Lộc	Leader	Locptse184400@fpt.edu.vn	0373097930
Trần Thái Thịnh	Member	Thinhttse182004@fpt.edu.vn	0338606577
Nguyễn Duy Khôi	Member	Khoindse184910@fpt.edu.vn	0783629758
Mai Thành Nhân	Member	Nhanmtse184916@fpt.edu.vn	0908554754

2. Product Background

The entrance examination preparation sector in Vietnam is experiencing substantial transformation driven by competency-based testing standards. The VNU-HCM Competency Assessment Exam (DGNL) has emerged as a crucial pathway for higher education admissions, requiring students to master complex interdisciplinary reasoning across linguistics, quantitative logic, and scientific problem-solving. As multi-branch test preparation centers expand across Ho Chi Minh City, traditional instructional models present significant limitations:

- **One-size-fits-all instruction:** Fixed classroom and static video curricula fail to accommodate variances in students' baseline abilities and target scores.
- **Inefficient learning focus:** Students spend excessive time reviewing mastered content while missing personalized remediation for foundational cognitive gaps.
- **Limited multi-campus synchronization:** Centers lack automated tools to balance self-paced digital learning with localized branch operations and live teacher-led interventions.
- **Lack of grounded, real-time guidance:** Generic question banks only grade raw scores without tracing skill mastery states or providing step-by-step Socratic explanations.

With the continuous growth of adaptive learning technologies and multi-campus educational demand, there is a strong need for an intelligent Online Blended Learning platform that personalizes learning paths, traces atomic competencies, and provides real-time AI tutor support for DGNL candidates.

3. Existing Systems

3.1 VETA EDU

- **Link:** https://veta.edu.vn
- **Description:** VETA EDU is a specialized online exam preparation platform focusing directly on the VNU-HCM Competency Assessment Examination (V-ACT / DGNL). It provides structured online courses, live-streamed coaching, and a dedicated online mock test room simulating the 120-question, 150-minute exam format with regional leaderboards.
- **System actors:**
    - Administrator: Manages student enrollments, exam packages, and class schedules.
    - Teacher: Delivers live online lectures, curates subject-specific test questions, and explains answer keys.
    - Student: Enrolls in targeted DGNL courses, takes timed mock tests, tracks leaderboard rankings, and reviews solution materials.
- **Features:**
    - Student: Takes standardized 120-question mock exams under countdown timers, submits responses, views overall scores and leaderboards, and accesses text/video answer explanations.
    - Teacher & Administrator: Schedules exam trials, uploads course lecture videos, and manages student test records.
- **Advantage and Disadvantage:**
    - Advantages: Highly specialized for the VNU-HCM DGNL structure; comprehensive question sets covering all three exam sections (Language, Logic/Math, Problem Solving); realistic online test room interface.
    - Disadvantages: Curricula and mock tests are statically delivered in fixed batches rather than dynamically customized; relies purely on raw scores without Item Response Theory (IRT) ability calibration or Bayesian Knowledge Tracing (BKT); lacks an integrated 24/7 AI Tutor to provide Socratic reasoning guidance; and cannot manage branch-level cohort operations across multiple physical campuses.

3.2 DOL THPT (DOL English)

- **Link:** https://dolthpt.vn/dgnl-va-dgtd
- **Description:** DOL THPT is an EdTech platform with an extensive campus footprint across Ho Chi Minh City, offering a technology-driven online practice testing environment with over 400 mock tests specifically tailored for the VNU-HCM DGNL examination.
- **System actors:**

    - System Administrator: Oversees question banks, digital platforms, and user management.
    - Teacher / Content Creator: Inputs exam questions, categorizes test types, and writes step-by-step solutions.
    - Student: Takes timed online practice tests, submits answers, reviews mistakes, and views practice histories.
- **Features:**
    - Student: Selects full-length or micro mock tests, completes online tests with automated timer countdowns, and reviews detailed explanations for each question.
    - System: Automatically grades submissions, computes total raw scores, and logs practice attempts over time.
- **Advantage and Disadvantage:**
    - Advantages: Strong brand presence across Ho Chi Minh City; modern and responsive testing interface; extensive test bank categorized by exam sections.
    - Disadvantages: Tests are treated as isolated exam papers without an underlying prerequisite skill graph (DAG); lacks adaptive sequencing (ZPD) to dynamically adjust difficulty to student competence; does not trace continuous knowledge mastery; and lacks interactive conversational AI tutoring support.

3.3 Tuyensinh247

- **Link:** https://on.tuyensinh247.com/thi-thu/ky-thi-dgnl-hcm.html
- **Description:** Tuyensinh247 is a major online exam preparation platform in Vietnam that operates a dedicated digital mock testing portal for high school candidates preparing for the VNU-HCM DGNL exam.
- **System actors:**
    - Administrator: Manages user accounts, course packages, and platform billing.
    - Teacher: Compiles exam questions, records video solutions, and hosts scheduled livestreams.
    - Student: Enrolls in DGNL courses, completes timed 150-minute mock tests, and reviews answer explanations.
- **Features:**
    - Administrator & Teacher: Organizes question banks by the 3 core DGNL domains (Language, Math/Logic, Problem Solving) and publishes timed mock examinations.
    - Student: Takes full-length mock exams with countdown timers, submits answers, checks scores, and reviews text and video solutions.
- **Advantage and Disadvantage:**
    - Advantages: Comprehensive question bank closely matching the official VNU-HCM DGNL matrix, high brand credibility, and realistic timed exam simulations.
    - Disadvantages: Curricula and study paths are static and linear without initial diagnostic profiling; only evaluates raw test scores without tracking atomic skill mastery states (IRT/BKT); lacks real-time difficulty adaptation; and does not support multi-campus center governance.

4. Business Opportunity

The VNU-HCM Competency Assessment Examination (DGNL) has become one of the main admission methods for universities in southern Vietnam, attracting over 130,000 registrations each year. As a result, test preparation centers in Ho Chi Minh City face increasing demand from 12th-grade students aiming for high target scores (such as 800+ or 900+ out of 1200).

However, existing local platforms (such as Tuyensinh247 or VETA EDU) mainly operate as static video portals or fixed test banks. Students purchase courses, watch the same pre-recorded lectures, and do 120-question mock tests under a timer. When students make mistakes, these systems only display raw scores and static text explanations. If students do not understand the solution, they have to post questions on Facebook groups or wait for weekly livestream sessions. More importantly, these platforms treat all learners the same; they do not diagnose which specific skill a student is missing (for example, reading comprehension vs. quantitative logic) or adapt question difficulty based on student ability.

Currently, there is no specialized EdTech platform in Ho Chi Minh City that combines multi-campus center management with adaptive learning algorithms (such as IRT and BKT) and conversational AI support.

Our project fills this market gap by delivering an online blended learning platform tailored specifically for the VNU-HCM DGNL structure. The platform gives students a personalized study roadmap with adaptive exercises that match their current ability level, while providing centers with centralized tools to manage branch operations, track class heatmaps, and assign teachers for targeted live Q&A sessions.

5. Software Product Vision

Our platform aims to transform VNU-HCM DGNL exam preparation through a unified online learning ecosystem that enables students to learn adaptively based on their personal abilities and empowers multi-campus test prep centers to operate more efficiently. Our vision is to create an intelligent, reliable, and accessible EdTech platform that personalizes learning paths, improves study efficiency, and supports both learners and educators.

6. Project Scope & Limitations

6.1 Major Features

Common User

- **FE-01:** Register & Verify Account (Register new account with mandatory email/OTP verification).
- **FE-02:** User Authentication (Login with role-based JWT access and refresh tokens).
- **FE-03:** Manage Profile & Security (View/update contact information and change password)

Student

- **FE-04:** Setup Goals & Select Campus (Configure target score on 1200 scale, exam date, daily study hours, and affiliated branch).

- **FE-05:** Take Diagnostic Assessment (Complete a 30-item test to calibrate initial latent ability theta_0).
- **FE-06:** Track Dynamic Learning Roadmap (Follow personalized milestones with lecture videos, quizzes, and Live Q&A schedules).
- **FE-07:** Adaptive Practice (Practice questions in ZPD with continuous mastery probability P(Lt) tracking via BKT).
- **FE-08:** Interact with AI Socratic Tutor (24/7 guided reasoning via SSE streaming backed by RAG from verified textbooks).
- **FE-09:** Join Virtual Classroom (Participate in real-time Live Q&A sessions hosted by campus teachers).
- **FE-10:** Access Learning Resources (Browse lecture videos and theoretical summaries mapped to competency skills).
- **FE-11:** Take Standardized Mock Exams (Complete 120-item timed mock tests with automated psychometric scoring).
- **FE-12:** View Analytics Dashboard & Score Prediction (Multi-dimensional radar charts, study streaks, and predicted university admission scores).
- **FE-13:** Generate Parent Invitation Code (Create secure token/QR code to grant parents access to study records).

Parent

- **FE-14:** Link Student Account (Redeem authorization token to link account with student profile).
- **FE-15:** Monitor Progress & Exam Readiness (Track completed study hours, milestone completion rates, and mastered skills).
- **FE-16:** Receive Early Warning Notifications (Automated alerts for consecutive 3-day inactivity or risk of missing target scores).
- Teacher
- **FE-17:** Virtual Classroom Management (Create virtual rooms, host Live Q&A sessions, and record attendance).
- **FE-18:** Moderate AI-Generated Content (Review, edit, and approve/reject AI-generated practice items and explanations).
- **FE-19:** Monitor Class Competency Heatmap (Supervise real-time mastery heatmaps to identify low-mastery clusters with $P(Lt) < 0.50$).
- **FE-20:** Pedagogical Intervention & Feedback (Provide evaluations and interventions for students flagged with learning difficulties).
- **FE-21:** Manage Campus Learning Materials (Upload supplementary exercises and reference documents mapped to skills).

Academic Manager

- **FE-22:** Allocate Teachers to Classes (Assign subject teachers to classes across managed campuses).
- **FE-23:** Approve Placement Cohorts (Review and confirm automated class placements derived from diagnostic tests).
- **FE-24:** Coordinate Schedules & Mock Exams (Manage schedules, facilities, and centralized offline mock exam sessions at campuses).
- **FE-25:** Supervise At-Risk Students (Track campus-wide lists of struggling students to enforce timely academic support).

Academic Director

- **FE-26:** Manage Master Competency Framework (Standardize and maintain the hierarchical competency graph structured as a DAG).
- **FE-27:** Administer Master Item Bank (Curate the master question bank and validate IRT 2PL psychometric parameters a, b).
- **FE-28:** Govern RAG Knowledge Base (Ingest, standardize, and index textbook materials into Vector DB for AI Tutor).
- **FE-29:** Tune Psychometric Algorithms (Calibrate domain weights, placement cutoff boundaries, and adaptive testing parameters).
- **FE-30:** Inter-Campus Analytics Dashboard (Benchmark training performance, score improvements, and curriculum quality across branches).

Administrator

- **FE-31:** Manage System Users & RBAC (Manage global accounts, assign roles, and handle ban/unban actions).
- **FE-32:** Manage Campus Directory (Provision and update the network of physical campus locations).
- **FE-33:** Configure System & AI Services (Manage system parameters, LLM API keys, token limits, and rate-limiting rules).
- **FE-34:** Data Backup & Recovery (Execute scheduled automated database backups and system disaster recovery).
- **FE-35:** Monitor System Health (Track server loads, service response latency, and system error logs).

6.2 Limitations

#	Limitations
LI-01	**Internet Dependency:** A stable internet connection is required for real-time testing, continuous BKT state tracking, SSE streaming chat, and virtual Live sessions. Extended network outages will interrupt active examination sessions.
LI-02	**Third-party LLM Dependency:** The AI Tutor relies on external LLM APIs (e.g., Google Gemini). Upstream service disruptions, token quota exhaustion, or rate limits will delay responses or trigger fallback to static solutions.
LI-03	**Vector DB Retrieval Limitation:** To eliminate hallucinations, the AI Tutor only answers if retrieved documents achieve Cosine Similarity ≥ 0.78 in Qdrant. Out-of-syllabus queries must be routed to instructor review queues.
LI-04	**BKT Model Convergence Delay:** The 30-item diagnostic test only provides a coarse initial ability estimate θ_0. The

	BKT model requires subsequent practice interactions to converge accurately upon true student mastery levels.
LI-05	**System Limitations (Requiring Pedagogical Intervention):** Automated adaptations cannot fully replace human pedagogy. Edge cases like persistent remedial bottlenecks (≥ 3 consecutive failures) or disputed questions mandate direct instructor intervention.
LI-06	**Localization & Curriculum Scope:** The platform is specifically calibrated for the VNU-HCM Competency Assessment Examination (ĐGNL ĐHQG-HCM) and currently operates only in Vietnamese.
LI-07	**OTP & Notification Service Dependency:** Account verification and emergency academic alerts to parents rely on third-party email/SMS gateways, where delivery delays may occasionally occur.
LI-08	**Cloud Storage & CDN Dependency:** Video lecture streaming and test diagram rendering depend directly on cloud storage availability and CDN network bandwidth.
LI-09	**Physical Campus Infrastructure Dependency:** Centralized offline mock examinations are constrained by workstation availability, room capacity, and proctoring personnel at individual campuses.

II. Project Management Plan

1. Overview

1.1 Scope & Estimation

#	WBS Item	Complexity	Est. Effort (man-days)
1	**Identity & Access Management (Authentication & RBAC)**		6.0

1.1	User Account Registration (Student / Parent)	Simple	1.0
1.2	Authentication & JWT Token Generation (Access & Refresh Token)	Medium	1.5
1.3	Account Activation via Email OTP Verification	Simple	1.0
1.4	Secure Password Reset & Password Change Management	Medium	1.5
1.5	Role-Based Access Control (RBAC) via JWT Claims	Simple	1.0
2	**Profile & Campus Directory Management**		6.0
2.1	View & Update User Profile Details	Simple	1.0
2.2	Configure Target Exam Score (1200 scale), Exam Date & Campus Selection	Medium	1.5
2.3	Generate Secure Parent-Student Authorization Pairing Code	Simple	1.0
2.4	Parent Account Linking via Student Verification Token	Simple	1.0
2.5	Campus Branch Directory Management (Campus CRUD	Medium	1.5
3	**Diagnostic Assessment & Automated Placement**		14.0

3.1	30-Item Abbreviated Diagnostic Test Generation Engine	Medium	2.0
3.2	Test Submission, Automated Scoring & Response Latency Logging	Medium	2.0
3.3	Baseline Ability Vector Estimation (θ) via 2PL IRT Model	Complex	3.5
3.4	Prior Knowledge Mastery Mapping P(L_0) via Clamped Logistic Sigmoid	Medium	2.0
3.5	Three-Tier Automated Cohort Placement & Campus Class Enrollment	Medium	2.5
3.6	Multi-Domain Competency Radar Chart Visualization	Medium	2.0
4	**Curriculum Framework & Master Item Bank Governance**		13.5
4.1	Competency Framework Management structured as a DAG	Complex	3.0
4.2	Cycle Detection in Competency Dependency Graph via Tarjan's Algorithm	Complex	3.0
4.3	Master Item Bank Governance & 2PL IRT Calibration (a, b)	Complex	3.5
4.4	Digital Learning Materials Management (Micro-lectures & Summaries)	Medium	2.0

5	**Personalized Learning Path Planning**		10.5
5.1	Milestone Sequence Extraction via Topological Sort Algorithm	Complex	3.5
5.2	Adaptive Path Pruning under Tight Exam Time Constraints	Complex	3.0
5.3	Integration of Micro-Lectures, Formative Quizzes & Live Q&A Schedules	Medium	2.0
5.4	Learning Milestone Unlock Condition & Progress State Tracking	Medium	2.0
6	**Adaptive Practice via ZPD & Bayesian Knowledge Tracing**		13.0
6.1	Item Recommendation Engine within ZPD Target Band (0.60 - 0.75)	Complex	4.0
6.2	Dynamic Knowledge State Updating via Bayesian Knowledge Tracing (BKT)	Complex	4.0
6.3	Milestone Mastery Completion Evaluation	Medium	2.0
6.4	Learning Bottleneck Detection & Remedial Node Insertion	Complex	3.0
7	**Socratic AI Tutor & Knowledge Retrieval Pipeline**		16.0
7.1	Curriculum Document Parsing, Chunking & Vector Ingestion into Qdrant	Complex	3.5

7.2	RAG Pipeline Retrieval & Strict Cosine Similarity Filtering (>= 0.78)	Complex	3.5
7.3	Real-Time Socratic Dialogue Engine via Server-Sent Events (SSE)	Complex	4.0
7.4	Isomorphic Formative Item Generation for Comprehension Checks	Complex	3.0
7.5	API Rate Limiting & Semantic Response Caching via Redis	Medium	2.0
8	**Virtual Classroom Operations & Standardized Mock Exams**		11.0
8.1	Virtual Classroom Scheduling, Live Q&A Hosting & Attendance Tracking	Complex	3.5
8.2	120-Item Standardized Timed Mock Examination Engine	Complex	4.0
8.3	Cohort-wide Score Equating & Standardization via Z-Score Normalization	Medium	2.0
8.4	Teacher Qualitative Feedback & Pedagogical Assessment Delivery	Medium	1.5
9	**Learning Analytics, Heatmaps & Early Warning System**		16
9.1	Student Performance Dashboard, Mastery Tracking & Study Streaks	Medium	2.5

9.2	Class Competency Heatmap with Deficit Zone Highlighting for Teachers	Complex	3.5
9.3	Parental Oversight Dashboard (Self-Study Hours & Skill Improvements)	Medium	2.0
9.4	Examination Score Range Projection (1200 Scale) & Target Probability	Complex	3.0
9.5	Multi-Channel Early Warning Engine (Push Notification, Email, Alerts)	Medium	2.5
9.6	Background Aggregation Job for Periodic Matrix Computation	Medium	2.5
10	**System Administration & Infrastructure Configuration**		6.5
10.1	System-wide User Account Governance (Activation, Suspension, Roles)	Medium	1.5
10.2	Psychometric Parameter Calibration & AI Token Quota Configuration	Medium	2.0
10.3	Infrastructure Health Monitoring (Server Load, Latency & Error Logs)	Medium	2.0
10.4	Automated Database Backup Scheduling & Disaster Recovery Protocols	Simple	1.0

Total Estimated Effort (man-days) **112.5 man-days**

1.2 Project Objectives

Overall Objective

#	Testing Stage	Test Coverage	No. of Defects	% of Defects	Notes
1	Reviewing	100% of SRS, architectural designs (ERD/C4), and service contracts (Proto/API)	10 – 20	~10%	Early defect detection through requirement analysis, database schema inspections, and inter-service contract walkthroughs. Mitigates architectural misalignment and reduces defect leakage into subsequent phases.
2	Unit Testing	≥ 80% source code coverage	50 – 60	~40%	Executed by developers on individual methods, classes, and business logic using automated test frameworks (xUnit, NUnit). Focuses on algorithmic accuracy, validation rules, and isolated domain logic. High defect discovery rate expected.
3	Integration Testing	≥ 70% interface and module interaction coverage	30 – 40	~25%	Validates data pipelines, inter-service gRPC communication, database persistence via ORM, and third-party integrations (e.g., Caching, Vector DB). Detects schema mismatches, data serialization errors, and interface integration faults.
4	System Testing	≥ 80% of total test cases executed	20 – 30	~20%	End-to-end validation of complete business workflows across all system modules via the API Gateway. Evaluates both functional requirements and non-functional attributes (security, stability, and response latency).
5	Acceptance Testing (UAT)	≥ 80% UAT scenarios executed	< 5	~5%	Validated by academic advisors and designated domain stakeholders within a staging environment simulating production conditions. Zero critical or blocker defects are permitted prior to capstone defense.
	Total		**110 – 155**	**100%**	

1.3 Project Risks

#	Risk Description	Impact	Likelihood	Risk Level	Response Plans

1	Security risks in handling user credentials and item bank leakage	High	Medium	High	Encrypt passwords using BCrypt/Argon2; enforce HTTPS and signed JWT authentication; mask correct answers and IRT psychometric parameters (a, b) at the API Gateway; perform scheduled automated vulnerability scans
2	Third-party LLM (Google Gemini) service outage, latency, or token quota exhaustion	High	Medium	High	Implement Circuit Breaker and automated retry policies; enforce strict rate-limiting (maximum 10 messages/minute/user); deploy Redis caching for semantically identical queries; fall back to static database solutions when the external AI API is unavailable.
3	Hallucination or pedagogical deviation from the AI Socratic Tutor	High	Medium	High	Enforce strict RAG similarity verification (Cosine Similarity >= 0.78 in Qdrant); apply multi-layered system prompt constraints against direct answer disclosure; route out-of-syllabus inquiries to teacher moderation queues.
4	Ambiguous or evolving requirements regarding competency exam structures	High	Medium	High	Enforce a formal Change Request (CR) protocol; freeze system scope following the analysis phase; standardize competency trees and item classification strictly against the VNU-HCM V-ACT framework.
5	System performance degradation under concurrent high load during Mock Examinations	Medium	High	Medium	Optimize SQL queries and index examination submission tables; utilize connection pooling across database contexts; execute comprehensive load testing (via k6/JMeter) prior to deployment

6	Curriculum graph degradation (cyclic dependency in competency DAG)	High	Low	Medium	Execute Tarjan's strongly connected components algorithm prior to Topological Sorting; abort graph operations and dispatch immediate alerts to the Academic Director upon cycle detection.
7	Third-party Email / OTP delivery failure or network queue latency	Medium	Medium	Medium	Implement a 60-second cooldown period between resend attempts; display transparent client-side status notices; store verification tokens in Redis with an explicit 5-minute Time-To-Live (TTL).
8	Qdrant Vector Database or Redis Cache service interruption	Medium	Low	Low	Configure automatic reconnection policies; allow temporary direct database fallback from PostgreSQL when Redis is down; record detailed operational error logs for administrator review
9	Inter-service gRPC communication failure (network timeout or proto contract mismatch)	High	Low	Medium	Enforce strict semantic versioning for all `.proto` files; configure explicit call deadlines and timeouts; centralize distributed request tracing and logging at the API Gateway to isolate service faults

2. Management Approach

2.1 Project Process

[Draw and describe the software development process model that the team would apply into the project implementation]

2.2 Quality Management

2.2.1 Defect Prevention

- **Coding Standards:** Establish and strictly enforce language-specific conventions for the .NET backend (C# Clean Architecture principles, static code analysis via Roslyn analyzers) and the mobile frontend (React Native with TypeScript linting rules, ESLint, and Prettier). Standardized naming conventions, modular architectural

separation, and automated formatting minimize common syntax errors while improving code readability and long-term maintainability.
- **Interface Contract Standardization:** Formalize inter-service communication definitions using gRPC `.proto` contracts and OpenAPI/Swagger schemas before initiating service implementation. This preventive measure eliminates data type discrepancies, unmapped payload fields, and breaking contract changes between distributed services.

2.2.2 Reviewing

- **Code Reviews:** Implement mandatory peer code reviews through Git Pull Requests (PRs) before merging code into the primary integration branches (`develop` and `main`). Each PR requires approval from at least one peer reviewer to verify adherence to architecture guidelines, algorithmic efficiency, security considerations, and early detection of potential logic flaws.
- **Design Reviews:** Conduct technical design walkthrough sessions prior to implementation in each sprint. These evaluations verify entity-relationship schemas (multi-schema PostgreSQL), graph integrity for competency structures (Direct Acyclic Graph validation), and ensure alignment between software architecture and formal SRS specifications.

2.2.3 Unit Testing

- **Component Validation:** Develop comprehensive unit test suites (using xUnit/NUnit for .NET services and Jest for React Native components) to validate individual methods, service handlers, and utility functions in complete isolation. Testing focuses primarily on core psychometric calculations (IRT 2PL parameter estimation, BKT mastery probability transitions), data validation pipelines, and critical business rules.
- **Automated Regression Prevention:** Integrate automated unit testing execution directly into the CI/CD pipeline. Tests execute automatically upon each pull request and commit, preventing regression bugs and ensuring that newly added features do not compromise existing system functionality.

2.2.4 Integration Testing

- **Module Interaction:** Execute integration tests across major development milestones to verify synchronous and asynchronous interactions between system components. This includes inter-service gRPC calls among the Identity, Practice, and Content services, as well as API Gateway route dispatching.
- **Data Flow Consistency:** Identify and resolve edge cases concerning data serialization, transactional consistency across PostgreSQL database contexts via Entity Framework Core, and persistence interactions with Redis cache and Qdrant vector database.

2.2.5 System Testing

- **Full End-to-End Validation:** Perform comprehensive black-box and end-to-end (E2E) testing covering the platform's core operational workflows—from student

diagnostic test assessment and automated classroom placement to adaptive practice sessions and timed mock exams.

- **Non-Functional Testing:** Evaluate critical non-functional criteria, including load performance under simulated high concurrent exam submissions (via k6/JMeter), role-based access security (JWT claims enforcement), and cross-platform mobile user interface responsiveness on React Native target devices.

2.2.6 Continuous Improvement

- **Process Refinement:** Re-evaluate and refine engineering and quality procedures following each Sprint Retrospective. Technical bottlenecks, integration roadblocks, and environment inconsistencies are documented alongside actionable remediation tasks for succeeding sprints.
- **Metrics-Driven Governance:** Utilize objective engineering metrics—including defect density across testing tiers, line and branch code coverage (targeting >= 80%), API response latency, and structured feedback from the academic advisor—to iteratively enhance software delivery and project quality.

2.3 Training Plan

Training Area	Participants	When, Duration	Waiver Criteria
Phát triển ứng dụng Mobile với React Native	(Điền tên bạn làm Mobile)	Tuần 1 - Tuần 2 (14 ngày)	Mandatory(Optional)

3. Project Deliverables

#	Deliverable	Due Date	Notes
1	Project Plan Document		Almost completing the project plan
2	Use Case Specification		Functional requirements across 6 user roles
3	Design Documents		UI/UX design, System architecture, and ERD
4	SRS Document		The final version of the Software Requirements Specification document
5	Web Admin & Management Modules		Web portal for Admin, Academic Manager, and Teacher
6	Mobile Client Application		Mobile app (React Native) for students and parents
7	Core Backend & AI Modules		.NET microservices, gRPC contracts, and IRT/BKT engine

8	Testing Documents		Test plan, test cases, and execution report

4. Responsibility Assignments

D ~ Do; R ~ Review; S ~ Support; I ~ Informed; \ ~ Omitted

Responsibility	locptse184400	thinhttse182004	khoindse184910	nhanmtse184916
Project Planning & Tracking	D	S	R	R
Prepare Project Introduction Document	D	S	R	R
Prepare SRS Document	R	S	D	R
Architecture Design Document	S	D	R	R
Database Design Document	S	D	R	R
Prepare Software Test Plan Document	R	S	R	D

5. Project Communications

Communication Item	Who/ Target	Purpose	When, Frequency	Type, Tool, Method(s)
Working with advisor	Supervisor (Nguyen Trong Tai)	- Review documentation - Demonstrate features - Evaluate progress and result	1–2 times per week	Face to face/Online meeting
Working with team	Team members	- Raise problems and blockers (Daily meeting) - Review source code - Evaluate progress and sprint results	Daily	Zalo, Discord, and Face to face

6. Configuration Management

6.1 Document Management

Manage projects with: Google Drive, Docs, Sheets, and Jira.

- **Drive Structure & Naming Conventions:**
  Organize project documents in a Google Drive folder with subfolders: Diagram, Topic, Report, Agenda/Note, Review.
  Documents should follow the format: [Title] - [Version] (e.g., Review 1).
- **Version Control:**
  Use Google Drive Version History to track document changes. Name major versions for easy identification.
- **Track Changes:**
  Enable "Suggest" mode for edits and add comments to get feedback.
  Ensure each change is recorded with revision details.
- **Ownership & Access:**
  Assign document owners to major documents to ensure appropriate accountability.
  Share documents with specific team members and set appropriate access rights (view, comment, edit).
- **Backup & Archive:**
  Regularly back up important documents.

6.2 Source Code Management

The project source code is managed using GitHub under the organization V-Eval, which contains the following repositories:

- System-Repo (Docker compose, deployment scripts)
- v-eval-gateway (API Gateway)
- v-eval-identity-service (Identity Service)
- v-eval-content-service (Content Service)
- v-eval-practice-service (Practice Service)
- v-eval-ai-engine (AI Engine & Psychometrics)

Branching Strategy:
Each repository is maintained under a task-based branch per developer model, designed to simplify collaboration while ensuring stability in the main branch:

- **Main branch:** the default and stable branch, used for deployment.
- **Dev branch:** integration branch where all developers merge and test their latest updates before main deployment.
- **Developer branches:** each team member maintains their own branch for handling assigned tasks.
  Developers work independently on their assigned branch. Once a task is completed and tested locally, changes are merged into dev via Pull Request. After the integration on dev passes testing, it is merged into main for production deployment.

Commit Convention:

- All commits follow the Conventional Commit standard to maintain consistent commit history and clear traceability:
  ():

Pull Request Workflow:

- Each developer works on their own personal branch (e.g., locpt, thinhtt, khoind, nhanmt).
- Once the feature or fix is completed, the developer creates a Pull Request (PR) from their personal branch into the dev branch.
- Team members review code before merging.
- Once the code in dev passes all functional tests, it is merged into main for deployment to the production server via GitHub Actions CI/CD.

Version Control & Deployment:

- Each repository is private, ensuring source code confidentiality.
- GitHub Actions automates backend deployment to a VPS using Docker Compose.
- Each merge to main represents a stable version.

Team Collaboration:

- Backend & Services team: members work on their own branches (locpt, thinhtt, khoind, nhanmt), working under the dev branch.
- The team maintains consistent synchronization and performs weekly merge reviews to prevent conflicts.

6.3 Tools & Infrastructures

Category	Tools / Infrastructure
Technology	.NET 8/9, ASP.NET Core Web API, gRPC, Python
Database	PostgreSQL
Vector DB & AI	Qdrant Vector DB, Google Gemini Flash 2.0
Caching	Redis (OTP storage, data caching)
IDEs/Editors	Visual Studio 2022, Visual Studio Code, Antigravity
Diagramming	Draw.io, Lucidchart
Documentation	Google Docs, Sheets, Slides, Google Drive
Version Control	GitHub (Source Code), Google Drive (Documents)
Deployment Server	VPS (Ubuntu), Docker, Docker Compose
Project Management	Jira (scheduling, tasks, issues)

III. Software Requirement Specification

1. Product Overview

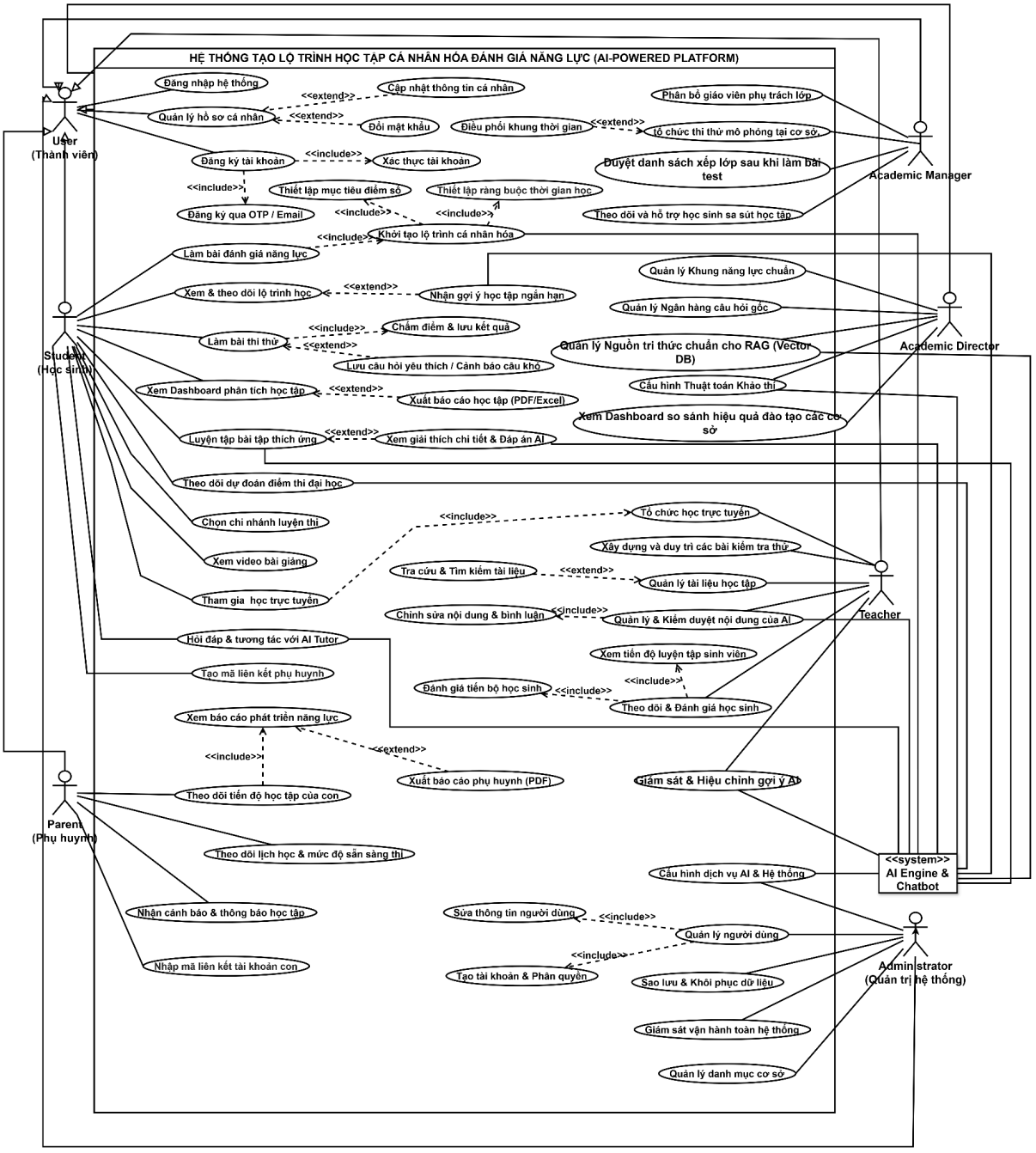
[This section presents a high-level overview of the product and the environment in which it will be used, the anticipated users, and known constraints, assumptions, and dependencies]

2. User Requirements

2.1. Actor:

Actor	Description
Guest	Unauthenticated user; browses platform introduction, VNUHCM-EPT exam structure, campus list, and takes sample trial tests.
Student	Enrolled learner; completes initial diagnostic tests (θ_0), studies personalized roadmaps, attends Live Q&A sessions, chats with AI Tutor, and tracks learning progress.
Parent	Student guardian; pairs accounts to monitor study hours, tracks Live Q&A attendance, and receives early progress alerts .
Teacher	Campus instructor; monitors class competency heatmaps, hosts real-time Live Q&A sessions, records attendance, and uploads recorded lecture videos.
Academic Manager	Campus academic head; manages class batches and teachers, coordinates mock exam schedules, approves placement lists, and handles skill dependency alerts.
Academic Director	Central academic authority; builds and approves the Master Taxonomy DAG, oversees master question banks, and configures psychometric parameters.
Administrator	Technical administrator; manages user accounts and RBAC, configures campus catalogs, monitors server infrastructure, and manages system parameters.

2.2. Use case diagram:



2.3. Use Case Description

ID	Use Case	Actors	Description
UC01	Register account	Guest	A guest can create a new account in the system by providing required personal information.

UC02	Verify account	Guest	A guest can verify their account via email OTP to activate access.
UC03	Login	All Users	A user can log into the system using email and password.
UC04	Logout	All Users	A user can log out of the system.
UC05	View profile	All Users	A user can view their personal profile information.
UC06	Update profile	All Users	A user can update their personal profile details.
UC07	Change password	All Users	A user can change their current account password.
UC08	Retrieve password	All Users	A user can reset a forgotten password via email verification.
UC09	Select campus branch	Student	A student can select a campus branch to enroll in classes and events.
UC10	Set target score & deadline	Student	A student can set their target exam score and official test date.
UC11	Take diagnostic test	Student	A student can take a diagnostic assessment to evaluate initial competencies.

UC12	Generate personalized roadmap	Student, AI System	The system can automatically create a personalized learning roadmap based on diagnostic results.
UC13	View roadmap timeline	Student	A student can view their learning roadmap milestones and progress status.
UC14	Receive study recommendations	Student, AI System	A student can receive short-term adaptive study suggestions from AI.
UC15	View video lectures	Student	A student can watch theory video lectures for each learning milestone.
UC16	Practice adaptive exercises	Student, AI System	A student can practice questions with difficulty automatically adjusted to their current level.
UC17	View detailed explanations	Student	A student can view detailed explanations and answers for practice questions.
UC18	Attend live class	Student	A student can attend real-time Live Q&A sessions hosted by teachers.
UC19	Watch recorded lecture	Student	A student can watch recorded videos of past live sessions.
UC20	Ask AI Tutor	Student, AI System	A student can chat with the Socratic AI Tutor to resolve exercise questions.
UC21	Take mock exam	Student	A student can take standardized mock exams under timed conditions.

UC22	View exam analytics	Student	A student can view detailed score breakdowns and results after taking an exam.
UC23	Bookmark questions	Student	A student can bookmark difficult or favorite questions for later review.
UC24	View analytics dashboard	Student	A student can view progress charts and multidimensional competency radar graphs.
UC25	Export study report	Student	A student can export their learning progress and test reports as PDF or Excel.
UC26	Track score predictions	Student, AI System	A student can track predicted exam score ranges and target attainment probabilities.
UC27	Create parent pairing code	Student	A student can generate a pairing code to link their account with a parent.
UC28	Enter student pairing code	Parent	A parent can enter a pairing code to link with their child's account.
UC29	Monitor learning progress	Parent	A parent can track their child's study hours and milestone completion.
UC30	View competency report	Parent	A parent can view competency radar reports and academic strengths/weaknesses.
UC31	Export parent report	Parent	A parent can export periodic study reports of their child in PDF format.

UC32	Receive notifications	Parent	A parent can receive automated notifications regarding their child's study alerts.
UC33	Track study schedule	Parent	A parent can track live class timetables and their child's exam readiness.
UC34	Host live class	Teacher	A teacher can host live online Q&A sessions and take attendance.
UC35	Build mock tests	Teacher	A teacher can build and schedule standardized mock tests for their class.
UC36	View learning materials	Teacher	A teacher can search and view instructional materials in the system repository.
UC37	Upload learning resources	Teacher	A teacher can upload documents, exercise files, and video lectures.
UC38	Update learning resources	Teacher	A teacher can edit or delete uploaded instructional materials.
UC39	View pending AI content	Teacher	A teacher can view AI-generated questions and rationales waiting for review.
UC40	Approve AI content	Teacher	A teacher can approve verified AI-generated questions into the official bank.
UC41	Edit AI content	Teacher	A teacher can edit AI-generated question items and explanations before approval.

UC42	View class heatmap	Teacher	A teacher can view the competency heatmap matrix of their assigned class.
UC43	Evaluate students	Teacher	A teacher can submit qualitative pedagogical feedback and evaluations for students.
UC44	Adjust AI suggestions	Teacher	A teacher can review and adjust AI-recommended study paths for students.
UC45	Assign teachers to classes	Academic Manager	An academic manager can assign teachers to online classes at their campus.
UC46	Coordinate timetable	Academic Manager	An academic manager can schedule live class time slots and local mock exam dates.
UC47	Approve class placement	Academic Manager	An academic manager can review and approve student class placement results.
UC48	Organize campus mock exam	Academic Manager	An academic manager can organize and oversee offline mock exams at the campus.
UC49	Monitor at-risk students	Academic Manager	An academic manager can track campus-wide student retention flags to coordinate support.
UC50	Manage competency taxonomy	Academic Director	An academic director can create and approve the standardized competency DAG.

UC51	Manage master question bank	Academic Director	An academic director can review and govern the central master question bank.
UC52	Manage RAG knowledge base	Academic Director	An academic director can upload and manage reference textbook embeddings in Vector DB.
UC53	Configure algorithm parameters	Academic Director	An academic director can configure IRT and BKT psychometric algorithm parameters.
UC54	View executive dashboard	Academic Director	An academic director can view comparative analytics across all campuses.
UC55	View accounts	Admin	An admin can view user accounts registered in the system.
UC56	Create account	Admin	An admin can create new staff accounts and assign system roles.
UC57	Update account	Admin	An admin can update user account information and roles.
UC58	Ban account	Admin	An admin can ban or unban user accounts violating policies.
UC59	Manage course categories	Admin	An admin can create, edit, and organize curriculum and course categories.
UC60	Manage campuses	Admin	An admin can create, update, and manage campus branch information.

UC61	Configure system settings	Admin	An admin can configure API keys, token quotas, and system parameters.
UC62	Monitor system health	Admin	An admin can monitor server resource loads and service availability.
UC63	Backup database	Admin	An admin can manage database backup and recovery operations.
UC64	Compute roadmap	AI System	The AI system can compute learning milestones using Topological Sort on the DAG.
UC65	Update student mastery	AI System	The AI system can calculate latent ability theta and update skill mastery states.
UC66	Predict exam scores	AI System	The AI system can forecast exam scores and trigger early risk notifications.

3. Functional Requirements

3.1 System Functional Overview

3.1.1. Screen flow:

3.1.1.1. Admin - Web Application Screen Flow

Figure 04. Admin - Web Application Screen Flow

3.1.1.2. Academic Director - Web Application Screen Flow

Figure 05. Academic Director - Web Application Screen Flow

3.1.1.3. Academic Manager - Web Application Screen Flow

Figure 06. Academic Manager - Web Application Screen Flow

3.1.1.4. Mentor & Teacher - Web Application Screen Flow

Figure 07. Mentor & Teacher - Web Application Screen Flow

3.1.1.5. Student - Web Application / Mobile Screen Flow

Web:
Figure 08. Student - Web Application Screen Flow

Mobile:
Figure 09. Student - Mobile Screen Flow

3.1.1.6. Parent - Portal Screen Flow

Figure 10. Parent - Portal Screen Flow

3.1.2. Screen description:

#	Feature	Screen	Description
SCR-00	Authentication	Welcome	Serves as the landing page for unauthenticated users, displaying the system branding, core value proposition of AI-personalized mentoring, and navigation options to Login and Registration.
SCR-01	Authentication	Register	Allows prospective students and parents to create an account by providing basic personal credentials: Full Name, Email, Phone Number, and Password, with mandatory acceptance of the Terms of Service.
SCR-02	Authentication	Login	Enables users to authenticate using Email/Password credentials or Google Single Sign-On (SSO). The system validates credentials and redirects users to their role-specific dashboards.
SCR-03	Authentication	Forgot password	Allows users to initiate account recovery by entering their registered email address to receive and verify a One-Time Password (OTP) before resetting their password.

SCR-04	Profile Management	Change Password	Enables authenticated users to update their credentials by validating their current password prior to saving a new password.
SCR-05	Profile Management	Manage Profile	Allows users to manage personal profile data, including Avatar, Contact Details, High School, and Grade 11–12 academic transcripts (used as reference context for Mentors). Registered email addresses remain read-only.
SCR-06	Information	HomePage	Displays examination schedules, available learning packages, featured mentors, system announcements, and user onboarding guidelines.
SCR-07	Chat	Chat	Facilitates real-time communication between Students and their assigned Mentors for academic inquiries and general counseling.
SCR-08	Chat AI	Socratic AI Tutor	Provides an interactive AI tutoring interface where students receive step-by-step guidance based on validated Question Patterns (Strict Grounding) without revealing answers directly. Includes an error reporting mechanism.
SCR-09	View Report	Report List	Displays a log of tickets and feedback submitted by users regarding incorrect questions, anomalous AI explanations, or technical issues.

SCR-10	View Report Detail	Report Detail	Displays the status and resolution history of a submitted report, permitting users to append supplementary information or media evidence.

Student

#	Feature	Screen	Description
SCR-11	Target Setting	Target Setting & Survey	Captures up to three target university majors during onboarding. The system automatically retrieves historical admission benchmark scores to establish the baseline Target Score and surveys home study hours.
SCR-12	Diagnostic Assessment	Diagnostic Test	Delivers a standardized 30-question diagnostic assessment mirroring the official VNUHCM-EPT section proportions. Captures item accuracy and response latency per question to evaluate initial competency (θ_0).
SCR-13	View Diagnostic Result	Competency Profile	Presents a multidimensional diagnostic report detailing initial proficiency, subject strengths/weaknesses, mastery per skill, and an estimated initial score range with Standard Error of Measurement (SEM).
SCR-14	Roadmap Management	Personalized Roadmap	Renders the adaptive Directed Acyclic Graph (DAG) learning roadmap, visualizing learning modules prioritized by Return on Investment (ROI) and their approval states (DRAFT or APPROVED).

SCR-15	Learning Stage	Learning Stage (P-L-A-R)	Guides students through the 4-step pedagogical loop: Quick Check (Preview), Concept/Pattern Video (Learn), Adaptive Practice (Apply), and Error Journaling (Reflect).
SCR-16	Practice	Adaptive Quiz	Delivers item-level adaptive practice driven by Bayesian Knowledge Tracing (BKT). Automatically schedules isomorphic review items for incorrect attempts via spaced repetition.
SCR-17	Mock Exam	Proctored Mock Exam	Delivers a 120-question, 150-minute simulated examination under controlled conditions, enforcing full-screen mode, tracking tab departures, and capturing periodic webcam snapshots for post-hoc proctoring.
SCR-18	Mistake Management	Mistake Notebook	Catalogs historical incorrect attempts categorized by Question Pattern, enriched with cognitive error tags assigned by Mentors during review sessions.
SCR-19	View Advisory	University Advisory	Displays 3-tier university admission recommendations (Reach, Target, Safety) generated by cross-referencing score forecast distributions against admission cut-off archives.
SCR-20	View Prediction	Score Forecast & What-If	Visualizes projected examination score distributions and actionable "What-If" simulations indicating target patterns required to attain specific score increments.

Parent

#	Feature	Screen	Description
SCR-21	Parent Survey	Parent Survey	A 3-minute intake survey capturing qualitative context regarding student self-discipline, home study routine, and psychological readiness.
SCR-22	View Child Progress	Child Progress Dashboard	Displays weekly academic engagement metrics, roadmap completion rates, and notifications regarding disciplinary escalation triggers.

Mentor & Teacher

#	Feature	Screen	Description
SCR-23	View Mentor Dashboard	Mentor Dashboard	Serves as the primary console for Lead Mentors to monitor assigned students, track progress, review early warning flags, and resolve assigned intervention tasks.
SCR-24	Roadmap Approval	Roadmap Kickoff & Approval	Interface for Mentors to review AI-generated DRAFT roadmaps, fine-tune milestones or parameters, and formally transition roadmaps to APPROVED status during kickoff consultations.

SCR-25	Cohort Management	Micro-cohort Management	Displays AI-clustered student micro-cohorts (≤ 6 students) sharing common pattern vulnerabilities, allowing manual mentor adjustments.
SCR-26	Mentoring Management	Mentoring Session	Manages bi-weekly masterclass workshops and 1-on-1/micro-coaching schedules. Automatically aggregates common incorrect questions across the cohort to streamline session prep.
SCR-27	Grading & Review	Exam Review & Proctoring Log	Allows Mentors to review student exam logs, assign qualitative cognitive error tags, and inspect suspicious activity flags from mock exam proctoring.
SCR-28	Early Warning Management	Intervention Task List	Tracks escalated student intervention workflows across the 3-tier disciplinary framework: push notification review, outbound check-in calls, and 3-way parent-mentor conferences.

Academic Director (Academic Quality & Content Governance)

#	Feature	Screen	Description

SCR-29	View Academic Dashboard	Academic Quality Dashboard	Provides macro-level governance over pedagogical assets, tracking item bank psychometric reliability, taxonomy coverage, and escalated content discrepancy rates.
SCR-30	Taxonomy Management	Pattern Taxonomy Management	Manages the 4-tier knowledge taxonomy (Domain → Subject → Skill → Archetype Pattern), including canonical solutions, distractor rationales, and common trap patterns.
SCR-31	Question Curation	Question Verification Queue	Serves as the gatekeeping queue to audit, calibrate, and approve ingested external examination items and AI-generated isomorphic questions into the canonical repository.
SCR-32	Academic Incident Review	Knowledge Error Resolution	Handles escalated student/mentor reports regarding academic inaccuracies, permitting direct overrides of canonical solution steps and vector knowledge groundings.

Academic Manager (Operations & Resource Allocation)

#	Feature	Screen	Description

SCR-33	View Operations Dashboard	Academic Operations Dashboard	Monitors operational metrics across all running cohorts, tracking syllabus adherence, overall attendance rates, mentor capacity utilization, and SLA compliance.
SCR-34	Resource Allocation	Mentor Assignment	Manages workload distribution and assigns Lead Mentors and Subject Mentors to incoming student cohorts, preventing mentor overallocation.
SCR-35	Benchmark Management	Admission Benchmark Config	Manages the repository of historical admission benchmark scores across university majors over the last 3 years to ensure accuracy for advisory models.
SCR-36	Escalation Monitoring	Tier-3 Intervention Case List	Oversees severe student disengagement cases escalating to Level 3, coordinating formal three-way conferences between parents, mentors, and academic management.

Admin (System Administration & Infrastructure)

#	Feature	Screen	Description
SCR-37	View Admin Dashboard	Admin Dashboard	Displays infrastructure-level telemetry, active user sessions across roles, background worker

			queues, and inference latency across internal microservices.
SCR-38	Account Management	Manage Accounts	Provides user identity administration across all system roles, supporting user provisioning, role assignments, suspension, and disciplinary status management.
SCR-39	System Report Management	System Issue Reports	Manages technical bug reports, platform exceptions, and service outages submitted by internal staff and end users.
SCR-40	System Config	System Configuration	Manages global algorithmic and runtime constants, including rapid-guessing thresholds ($< 5s$), item-stuck boundaries ($> 3m$), BKT mastery criteria (0.85), API credentials, and notification limits.

3.1.3. Screen authorization:

Screen	Admin	Academic Director	Academic Manager	Mentor / Teacher	Student	Parent	Guest
Welcome	x	x	x	x	x	x	x
Register	x	x	x	x	x	x	x
Login	x	x	x	x	x	x	x

Forgot password	x	x	x	x	x	x	x
Change Password	x	x	x	x	x	x	
Manage Profile	x	x	x	x	x	x	
HomePage	x	x	x	x	x	x	x
Chat				x	x		
Socratic AI Tutor				x	x		
Report List	x	x	x	x	x	x	
Report Detail	x	x	x	x	x	x	
Target Setting & Survey					x		
Diagnostic Test					x		
Competency Profile				x	x		

Personalized Roadmap				x	x	x	
Learning Stage (P-L-A-R)				x	x		
Adaptive Quiz					x		
Proctored Mock Exam				x	x		
Mistake Notebook				x	x		
University Advisory				x	x	x	
Score Forecast & What-If				x	x	x	
Parent Survey						x	
Child Progress Dashboard						x	

Mentor Dashboard				x			
Roadmap Kickoff & Approval				x			
Micro-cohort Management			x	x			
Mentoring Session			x	x			
Exam Review & Proctoring Log				x			
Intervention Task List				x			
Academic Quality Dashboard		x					
Pattern Taxonomy Management		x					
Question Verification Queue		x					

Knowledge Error Resolution		x					
Academic Operations Dashboard			x				
Mentor Assignment			x				
Admission Benchmark Config			x				
Tier-3 Intervention Case List			x				
Admin Dashboard	x						
Manage Accounts	x						
System Issue Reports	x						
System Configuration	x						

3.1.4. Non-screen functions:

#	Feature	System Function	Description
1	**Adaptive Learning Engine**	Real-time BKT Mastery Updater	Asynchronously consumes student item submission events via RabbitMQ to compute updated latent mastery probabilities $P(L_t)$ per skill/pattern in real time, determining next-item selection.
2	**Spaced Repetition Scheduler**	Isomorphic Review Job	Executes scheduled batch sweeps identifying failed items or concepts below mastery ($Mastery < 0.85$) to inject isomorphic practice variants into upcoming daily revision queues.
3	**Student Engagement & Escalation**	Inactivity & Anomaly Escalation Service	Inspects student activity telemetry daily to automate the 3-tier escalation framework: triggering push alerts (1–2 days inactive), assigning Lead Mentor outreach tasks (3 days), and scheduling 3-way conferences (5 days).
4	**Cohort Clustering Engine**	Micro-cohort Grouping Service	Executes periodic K-Means clustering over multidimensional pattern deficiency vectors, partitioning students within a cohort into micro-groups of ≤ 6 peers for targeted mentoring.

5	**Mock Exam Integrity Verifier**	Exam Proctoring Snapshot Analyzer	Ingests tab blur logs and periodic webcam snapshots captured during mock exams to calculate an aggregate integrity suspicion index, flagging suspect attempts for mentor review.
6	**Score Prediction Calibration**	Score Forecast Calibration Service	Periodically re-executes embedded ONNX regression models against empirical mock performance and roadmap velocity, refreshing projected score distributions and admissions recommendations.

3.1.5. Conceptual ERD:

Figure 11. Conceptual ERD

3.2 Authentication & Account Management

3.2.1 Login

- **Function trigger**: User clicks “Đăng nhập” button
- **Function description**:
    - **Actors**: Guest, Student, Parent, Mentor, Teacher, Academic Manager, Academic Director, Admin
    - **Purpose**: Authenticate the user to access authorized system features
- **Function details**:
    - **Data requirement**: email, password
    - **Validation**:
        - **email**:
            - Type: Email
            - Required
            - Condition: Must exist in the system
        - **password**:
            - Type: Password
            - Required
            - Min length: 8 characters
            - Must contain at least:
                - 1 uppercase letter
                - 1 lowercase letter
                - 1 digit
                - 1 special character
        - **account status**:
            - Condition: Account status must be ACTIVE

        - **Business rules**: BR-01, BR-02

3.2.2 Logout

- **Function trigger**: User clicks “Đăng xuất” button
- **Function description**:
    - **Actors**: All Authenticated Users
    - **Purpose**: End the current user session and revoke refresh tokens
- **Function details**:
    - **Data requirement**: None
    - **Validation**:
        - User has an active login session
    - **Business rules**: None

3.2.3 Forgot Password / Reset

- **Function trigger**: User clicks “Quên mật khẩu” button/link
- **Function description**:
    - **Actors**: Guest, All Users
    - **Purpose**: Initiate password recovery via OTP sent to the registered email
- **Function details**:
    - **Data requirement**: email
    - **Validation**:
        - **email**:
            - Type: Email
            - Required
            - Condition: Must exist in the system
    - **Business rules**: BR-01

3.2.4 Register Account

- **Function trigger**: User clicks “Đăng ký” button
- **Function description**:
    - **Actors**: Guest
    - **Purpose**: Create a new account in the system for prospective students or parents
- **Function details**:
    - **Data requirement**: email, password, confirmPassword, fullName, phone, termsAccepted
    - **Validation**:
        - **email**:
            - Type: Email
            - Required
            - Must follow valid email format
            - Must be unique in the system
        - **password**:
            - Type: Password
            - Required
            - Min length: 8 characters
            - Must contain at least:
                - 1 uppercase letter
                - 1 lowercase letter

                - 1 digit
                - 1 special character
            - **confirmPassword**:
                - Type: Password
                - Required
                - Must match password
            - **fullName**:
                - Type: Text
                - Required
                - Min length: 2 characters
                - Max length: 100 characters
            - **phone**:
                - Type: Phone number
                - Required
                - Must follow 10-digit Vietnamese phone format
                - Must be unique in the system
            - **termsAccepted**:
                - Type: Checkbox
                - Required
                - Must be checked
        - **Business rules**: BR-01, BR-03, BR-04

3.2.5 Verify Account OTP

- **Function trigger**: User clicks “Xác thực tài khoản” button
- **Function description**:
    - **Actors**: Guest
    - **Purpose**: Verify email ownership via a 6-digit OTP code to activate the account
- **Function details**:
    - **Data requirement**: email, otpCode
    - **Validation**:
        - **email**:
            - Type: Email
            - Required
            - Condition: Must exist in inactive status
        - **otpCode**:
            - Type: Numeric string
            - Required
            - Length: 6 digits
            - Condition: Valid within 10 minutes and unused
    - **Business rules**: BR-01, BR-02

3.2.6 Manage Profile

- **Function trigger**: User clicks “Lưu thay đổi” button on the profile page
- **Function description**:
    - **Actors**: All Authenticated Users
    - **Purpose**: Update personal contact details, school background, and daily study availability
- **Function details**:
    - **Data requirement**: fullName, phone, schoolName, studyHoursDay, campusId

- **Validation**:
  - **fullName**:
    - Type: Text
    - Required
    - Length: 2 to 100 characters
  - **phone**:
    - Type: Phone number
    - Required
    - Valid phone format
  - **studyHoursDay**:
    - Type: Numeric
    - Range: 0.5 to 12.0 hours/day
  - **campusId**:
    - Type: UUID
    - Condition: Must exist and be active
- **Business rules**: BR-04

3.2.7 Change Password

- **Function trigger**: User clicks “Đổi mật khẩu” button
- **Function description**:
  - **Actors**: All Authenticated Users
  - **Purpose**: Allow users to update their account password
- **Function details**:
  - **Data requirement**: currentPassword, newPassword, confirmNewPassword
  - **Validation**:
    - **currentPassword**:
      - Type: Password
      - Required
      - Condition: Must match existing password
    - **newPassword**:
      - Type: Password
      - Required
      - Min length: 8 characters
      - Must contain at least:
        - 1 uppercase letter
        - 1 lowercase letter
        - 1 digit
        - 1 special character
      - Condition: Must not match currentPassword
    - **confirmNewPassword**:
      - Type: Password
      - Required
      - Condition: Must match newPassword
  - **Business rules**: BR-03

3.3 Student Learning & Practice

3.3.1 Target Setting & Survey

- **Function trigger**: User clicks “Lưu nguyện vọng mục tiêu” button
- **Function description**:

    - **Actors**: Student
    - **Purpose**: Register target university majors, baseline target score, and self-study availability
- **Function details**:
    - **Data requirement**: targetMajors, targetScore, examDate, weeklyStudyHours
    - **Validation**:
        - **targetMajors**:
            - Required
            - Min 1, max 3 valid majors
        - **targetScore**:
            - Type: Integer
            - Required
            - Range: 300 to 1200
        - **examDate**:
            - Type: Date
            - Required
            - Condition: Must be at least 14 days in the future
        - **weeklyStudyHours**:
            - Type: Numeric
            - Required
            - Range: 5 to 50 hours/week
    - **Business rules**: BR-07

3.3.2 Take Diagnostic Test

- **Function trigger**: User clicks “Bắt đầu làm bài kiểm tra năng lực” button
- **Function description**:
    - **Actors**: Student
    - **Purpose**: Complete a 30-item assessment to evaluate initial trait level and identify skill gaps
- **Function details**:
    - **Data requirement**: diagnosticTestId, answers (30 items with selectedOption and timeSpentSeconds)
    - **Validation**:
        - **diagnosticTestId**:
            - Required
        - **answers**:
            - Required
            - Count: Exactly 30 questions matching official exam distribution
        - **timeSpentSeconds**:
            - Type: Integer
            - Required per item
            - Latencies under 5s flagged for discrimination parameter reduction
    - **Business rules**: BR-05, BR-06, BR-11, BR-13

3.3.3 View Competency Profile

- **Function trigger**: Automatically displayed upon diagnostic test submission or user clicks “Hồ sơ năng lực” tab
- **Function description**:
    - **Actors**: Student, Mentor

    - **Purpose**: Present diagnostic results including initial latent ability, SEM error, radar chart, and skill mastery levels
- **Function details**:
    - **Data requirement**: submissionId
    - **Validation**:
        - **submissionId**:
            - Required
            - Test submission must be fully evaluated
    - **Business rules**: BR-06

3.3.4 Generate Personalized Roadmap

- **Function trigger**: System-triggered automatically upon diagnostic evaluation completion
- **Function description**:
    - **Actors**: Student, AI System
    - **Purpose**: Compute and provision a personalized DAG-based learning roadmap in DRAFT state
- **Function details**:
    - **Data requirement**: studentId, targetScore, weakSkillList, initialTheta
    - **Validation**:
        - Skill DAG must be strictly acyclic
        - Skills with initial mastery >= 0.85 must be pruned
        - Initial roadmap status must be set to DRAFT
    - **Business rules**: BR-08, BR-09, BR-14

3.3.5 View Roadmap & Stages

- **Function trigger**: User clicks “Lộ trình học tập” menu
- **Function description**:
    - **Actors**: Student, Parent, Mentor
    - **Purpose**: Display stage progression, milestone completion percentages, and lock/unlock node statuses
- **Function details**:
    - **Data requirement**: studentId
    - **Validation**:
        - Student must have an initialized roadmap record
    - **Business rules**: BR-08

3.3.6 Learning Stage (P-L-A-R Cycle)

- **Function trigger**: User clicks an unlocked stage on the roadmap view
- **Function description**:
    - **Actors**: Student
    - **Purpose**: Guide the student through the 4-step sequence: Preview, Learn, Apply, and Reflect
- **Function details**:
    - **Data requirement**: stageId, studentId, stepSequence (1 to 4)
    - **Validation**:
        - **stageId**:
            - Required
            - Status must be IN_PROGRESS
        - **videoWatchRate**:
            - Completion rate must be >= 80% to unlock the quiz step

        - **Business rules**: BR-09, BR-14

3.3.7 Adaptive Quiz Session

- **Function trigger**: User clicks “Bắt đầu luyện tập thích ứng” button
- **Function description**:
    - **Actors**: Student, AI System
    - **Purpose**: Deliver adaptive quiz items, update BKT mastery parameters, and queue spaced repetition reviews upon failure
- **Function details**:
    - **Data requirement**: stageId, skillId, patternId, selectedAnswers
    - **Validation**:
        - **skillId, patternId :**
            - Required
            - Must belong to the active stage
        - **quizScore**:
            - Must achieve >= 60% and BKT mastery >= 0.85 for node completion
    - **Business rules**: BR-10, BR-14, BR-15

3.3.8 Socratic AI Tutor

- **Function trigger**: User sends a question in the AI Chat window
- **Function description**:
    - **Actors**: Student, AI System
    - **Purpose**: Provide 24/7 Socratic scaffolding grounded in verified pattern templates without revealing answers directly
- **Function details**:
    - **Data requirement**: questionId (optional), userMessage, conversationHistory
    - **Validation**:
        - **userMessage**:
            - Type: Text
            - Required
            - Max length: 1000 characters
            - Must pass system profanity and external contact filters
        - **grounding**:
            - Explanations must strictly adhere to validated canonical solutions
    - **Business rules**: BR-12, BR-13, BR-25

3.3.9 Proctored Mock Exam

- **Function trigger**: User clicks “Vào phòng thi thử” button
- **Function description**:
    - **Actors**: Student
    - **Purpose**: Deliver a full 120-item / 150-minute exam simulation with full-screen enforcement and random webcam integrity captures
- **Function details**:
    - **Data requirement**: examId, answers, proctoringSnapshots, tabDepartureEvents
    - **Validation**:
        - **examId**:
            - Required

            - Exam status must be published (IsPublished = true)
        - **fullScreen**:
            - Enforced full-screen display
            - More than 3 tab/window departures triggers an integrity violation flag
    - **Business rules**: BR-11, BR-21, BR-22

3.3.10 Mistake Notebook

- **Function trigger**: User clicks “Sổ tay lỗi sai” menu
- **Function description**:
    - **Actors**: Student, Mentor
    - **Purpose**: Catalog historical failed questions categorized by pattern and display mentor-assigned cognitive error tags
- **Function details**:
    - **Data requirement**: studentId, domainFilter, patternFilter
    - **Validation**:
        - Must display only items belonging to the student's attempt history
    - **Business rules**: BR-09, BR-15

3.3.11 University Advisory

- **Function trigger**: User clicks “Tư vấn nguyện vọng” tab
- **Function description**:
    - **Actors**: Student, Parent, Mentor
    - **Purpose**: Categorize university majors into Reach, Target, and Safety tiers based on score forecast intervals
- **Function details**:
    - **Data requirement**: studentId, targetMajors
    - **Validation**:
        - Benchmark evaluations must map to historical cut-off databases
    - **Business rules**: BR-23

3.3.12 Score Forecast & What-If

- **Function trigger**: User clicks “Dự báo điểm số” menu
- **Function description**:
    - **Actors**: Student, Parent, Mentor
    - **Purpose**: Display projected score ranges and actionable What-If target simulations
- **Function details**:
    - **Data requirement**: studentId
    - **Validation**:
        - Student must have completed either the diagnostic test or at least one mock exam
    - **Business rules**: BR-06, BR-23

3.3.13 Bookmark Questions

Chưa triển khai

3.3.14 Export Study Report

Chưa triển khai

3.4 Parent Portal

3.4.1 Pair Student Account

Chưa triển khai

3.4.2 Parent Survey

- **Function trigger**: User clicks “Gửi khảo sát” button on the 3-minute intake form
- **Function description**:
    - **Actors**: Parent
    - **Purpose**: Capture qualitative context regarding home study conditions, student discipline, and expectations
- **Function details**:
    - **Data requirement**: studentId, homeStudyEnvironment, selfDisciplineLevel, expectationScore
    - **Validation**:
        - **studentId**:
            - Required
            - Must have an active guardian link
        - **homeStudyEnvironment**:
            - Required, scale 1 to 5
        - **selfDisciplineLevel**:
            - Required, scale 1 to 5
    - **Business rules**: BR-07

3.4.3 View Child Progress Dashboard

- **Function trigger**: User navigates to the Parent Portal dashboard
- **Function description**:
    - **Actors**: Parent
    - **Purpose**: Monitor weekly roadmap progress, study streak continuity, and assessment score summaries
- **Function details**:
    - **Data requirement**: studentId
    - **Validation**:
        - Parent account must be actively linked to the student
    - **Business rules**: None

3.4.4 Receive Inactivity & Escalation Alerts

- **Function trigger**: System-triggered when student engagement falls below disciplinary thresholds
- **Function description**:
    - **Actors**: Parent
    - **Purpose**: Deliver automated push/email notifications regarding study inactivity and escalation conferences
- **Function details**:
    - **Data requirement**: notificationPayload, escalationTier (Tier 1, 2, or 3)
    - **Validation**:
        - Triggered strictly when inactivity crosses predefined threshold days
    - **Business rules**: BR-18, BR-19, BR-20

3.5 Teacher & Mentoring Operations

3.5.1 Mentor Cohort Dashboard

- **Function trigger**: User logs in and navigates to the Cohort Dashboard
- **Function description**:
    - **Actors**: Mentor, Teacher
    - **Purpose**: Provide visibility over assigned students, progress rates, traffic-light flags, and action items
- **Function details**:
    - **Data requirement**: mentorId, campusId
    - **Validation**:
        - Must display only students allocated to the logged-in mentor
    - **Business rules**: BR-16

3.5.2 Roadmap Kickoff & Approval

- **Function trigger**: User clicks “Phê duyệt lộ trình” button during the kickoff session
- **Function description**:
    - **Actors**: Mentor
    - **Purpose**: Review AI-generated draft roadmaps, adjust target allocations, and officially activate the plan
- **Function details**:

    - **Data requirement**: roadmapId, studentId, adjustmentNotes, approvedMilestones
    - **Validation**:
        - **roadmapId**:
            - Required
            - Roadmap status must currently be DRAFT
        - **mentorPermission**:
            - Only the designated Lead Mentor can approve
    - **Business rules**: BR-08

3.5.3 Micro-cohort Management

- **Function trigger**: User clicks “Quản lý nhóm chuyên đề” menu
- **Function description**:
    - **Actors**: Mentor, Academic Manager
    - **Purpose**: Manage AI-clustered student groups (<= 6 students) sharing common pattern vulnerabilities
- **Function details**:
    - **Data requirement**: cohortId, studentIdsList, targetPatternId
    - **Validation**:
        - Group membership must not exceed 6 students
    - **Business rules**: BR-17

3.5.4 Mentoring Session Management

- **Function trigger**: User clicks “Lên lịch buổi học” button
- **Function description**:
    - **Actors**: Mentor, Teacher
    - **Purpose**: Schedule workshops and micro-coaching sessions with aggregated shared student mistake items
- **Function details**:
    - **Data requirement**: sessionId, scheduledAt, meetingUrl, cohortId, attendanceList
    - **Validation**:

        - **scheduledAt**:
            - Must be a future timestamp
        - **meetingUrl**:
            - Must follow valid URL format (Zoom, Google Meet, MS Teams)
    - **Business rules**: BR-16, BR-17, BR-25

3.5.5 Exam Review & Cognitive Error Tagging

- **Function trigger**: User opens a student's completed exam attempt
- **Function description**:
    - **Actors**: Mentor, Teacher
    - **Purpose**: Review item responses, inspect proctoring flags, and assign cognitive error tags to mistakes
- **Function details**:
    - **Data requirement**: submissionId, questionId, errorTag, mentorComment
    - **Validation**:
        - **submissionId**:
            - Required
            - Exam status must be completed
        - **errorTag**:
            - Must select from standardized taxonomy tags
    - **Business rules**: BR-07, BR-21, BR-22

3.5.6 Intervention Task List (Disciplinary Escalation)

- **Function trigger**: User clicks “Trung tâm cảnh báo” tab
- **Function description**:
    - **Actors**: Mentor
    - **Purpose**: Resolve assigned check-in tasks for 3-day inactive students or 3-way conference cases
- **Function details**:
    - **Data requirement**: taskId, studentId, interventionLevel, resolutionNotes, status
    - **Validation**:
        - **taskId**:
            - Required
            - Notes are required upon marking status as RESOLVED
    - **Business rules**: BR-19, BR-20, BR-24

3.6 Academic Manager Operations

3.6.1 Mentor Workload & Assignment

- **Function trigger**: User clicks “Phân công Mentor” button
- **Function description**:
    - **Actors**: Academic Manager
    - **Purpose**: Allocate Lead Mentors and Subject Teachers to cohorts while enforcing workload caps
- **Function details**:
    - **Data requirement**: mentorId, studentIdsList, campusId
    - **Validation**:
        - **mentorId**:
            - Must be an active faculty user
        - **capacityCheck**:

            - Total active students assigned to a Lead Mentor must not exceed 20
    - **Business rules**: BR-16

3.6.2 Automated Class Placement Approval (K-Means)

- **Function trigger**: User clicks “Xếp lớp tự động” button
- **Function description**:
    - **Actors**: Academic Manager
    - **Purpose**: Approve automatic class clustering based on 4-domain deficiency vectors via K-Means
- **Function details**:
    - **Data requirement**: campusId, targetSubject, clusterList
    - **Validation**:
        - Clustered group sizes must satisfy cohort size thresholds
    - **Business rules**: BR-17

3.6.3 Admission Benchmark Configuration

- **Function trigger**: User clicks “Cập nhật điểm chuẩn” button
- **Function description**:
    - **Actors**: Academic Manager, Admin
    - **Purpose**: Manage the 3-year historical cut-off database across university majors for advisory features
- **Function details**:
    - **Data requirement**: universityCode, majorCode, majorName, benchmarkScore, academicYear
    - **Validation**:
        - **benchmarkScore**:
            - Integer from 300 to 1200
        - **academicYear**:
            - Valid past year (e.g., 2023, 2024, 2025)
        - **uniqueConstraint**:
            - Combination of universityCode + majorCode + academicYear must be unique
    - **Business rules**: BR-23

3.6.4 Tier-3 Escalation Case Management

- **Function trigger**: User opens the Tier-3 Escalation view
- **Function description**:
    - **Actors**: Academic Manager
    - **Purpose**: Coordinate formal 3-way conferences between parent, student, and mentor for chronic disengagement
- **Function details**:
    - **Data requirement**: studentId, mentorId, parentContact, meetingSchedule, actionPlan
    - **Validation**:
        - Student must have >= 5 consecutive inactive days
    - **Business rules**: BR-20

3.6.5 Academic Operations Dashboard

- **Function trigger**: User navigates to the Operations Dashboard
- **Function description**:
    - **Actors**: Academic Manager

        - **Purpose**: Track overall syllabus pacing, attendance metrics, and mentor SLA compliance
- **Function details**:
    - **Data requirement**: campusId, dateRange
    - **Validation**:
        - **campusId**:
            - Required and within user jurisdiction
    - **Business rules**: None

3.7 Academic Director Governance

3.7.1 Pattern Taxonomy Management

- **Function trigger**: User clicks “Thêm dạng bài mới” or “Lưu chỉnh sửa” button
- **Function description**:
    - **Actors**: Academic Director
    - **Purpose**: Maintain the 4-tier taxonomy (Domain -> Subject -> Skill -> Pattern), canonical solutions, and traps
- **Function details**:
    - **Data requirement**: domainCode, subjectCode, skillCode, patternCode, patternName, canonicalSolutionTemplate, commonTrapNotes
    - **Validation**:
        - Knowledge DAG graph must remain strictly acyclic
        - Every bank question must map strictly to a single leaf pattern
    - **Business rules**: BR-09

3.7.2 Question Verification Queue

- **Function trigger**: User clicks “Phê duyệt câu hỏi” button in the review queue
- **Function description**:
    - **Actors**: Academic Director
    - **Purpose**: Audit external ingested items or AI variants, transitioning status from PRACTICE_ONLY to canonical
- **Function details**:
    - **Data requirement**: questionId, reviewDecision (APPROVE, REJECT, REQUEST_EDIT), taxonomyTag, editedLatexContent
    - **Validation**:
        - **questionId**:
            - Required
        - **completeness**:
            - Must include 4 options and canonical explanation
        - Only verified items can be injected into diagnostic tests and mock exams
    - **Business rules**: BR-10, BR-11, BR-13

3.7.3 Knowledge Error Resolution (Socratic Guardrails)

- **Function trigger**: User opens an escalated content inaccuracy report
- **Function description**:
    - **Actors**: Academic Director
    - **Purpose**: Resolve reported pedagogical errors, override canonical solutions, and adjust RAG vectors
- **Function details**:

    - **Data requirement**: reportId, questionId, correctedSolution, resolutionNotes
    - **Validation**:
        - Corrected keys remain immutable once republished
    - **Business rules**: BR-12, BR-13

3.7.4 Configure Psychometric & Algorithm Parameters

- **Function trigger**: User clicks “Lưu cấu hình thuật toán” button
- **Function description**:
    - **Actors**: Academic Director, Admin
    - **Purpose**: Calibrate parameters for IRT 2PL, BKT priors, mastery thresholds, and rapid-guess filters
- **Function details**:
    - **Data requirement**: rapidGuessThresholdSeconds, masteryThresholdProbability, irtPriorMean, irtPriorVariance
    - **Validation**:
        - **masteryThresholdProbability**:
            - Float between 0.70 and 0.95 (default 0.85)
        - **rapidGuessThresholdSeconds**:
            - Integer between 3 and 10 seconds (default 5s)
    - **Business rules**: BR-06, BR-14

3.8 System Administration

3.8.1 Manage Accounts

- **Function trigger**: User accesses the User Administration console
- **Function description**:
    - **Actors**: Admin
    - **Purpose**: Manage user identities, provision staff accounts, and assign role access levels
- **Function details**:
    - **Data requirement**: userId, email, fullName, role, campusId, status (ACTIVE, SUSPENDED, BANNED)
    - **Validation**:
        - **email, phone**:
            - Unique across the platform
        - **role**:
            - Must match an authorized system role
    - **Business rules**: BR-02, BR-03, BR-04

3.8.2 Lock / Unlock Account (Ban Account)

- **Function trigger**: User clicks “Khóa tài khoản” or “Mở khóa tài khoản” button
- **Function description**:
    - **Actors**: Admin
    - **Purpose**: Suspend violating accounts or restore access following review
- **Function details**:
    - **Data requirement**: userId, banReason, action (LOCK, UNLOCK)
    - **Validation**:
        - **userId**:
            - Must exist in the database
        - **banReason**:

            - Mandatory when locking an account
        - All active refresh tokens are revoked upon suspension
    - **Business rules**: BR-02

3.8.3 Manage Campuses

- **Function trigger**: User clicks “Thêm cơ sở mới” or “Lưu thông tin cơ sở” button
- **Function description**:
    - **Actors**: Admin
    - **Purpose**: Manage the network of physical campus branches across regions
- **Function details**:
    - **Data requirement**: campusCode, campusName, address, contactPhone, isActive
    - **Validation**:
        - **campusCode**:
            - Type: Text, Required, unique
        - **campusName**:
            - Type: Text, Required
    - **Business rules**: None

3.8.4 System Configuration & API Management

- **Function trigger**: User clicks “Lưu thiết lập hệ thống” button
- **Function description**:
    - **Actors**: Admin
    - **Purpose**: Configure third-party API credentials, payment endpoints, and messaging integrations
- **Function details**:
    - **Data requirement**: configKey, configValue, description
    - **Validation**:
        - **configKey**:
            - Required and unique across settings
            -
    - **Business rules**: None

3.8.5 System Health & Audit Logs

- **Function trigger**: User accesses the Infrastructure Monitoring console
- **Function description**:
    - **Actors**: Admin
    - **Purpose**: Monitor service availability, container resource utilization, and system error logs
- **Function details**:
    - **Data requirement**: serviceName, timeRange, logSeverity
    - **Validation**:
        - Audit log records retained for at least 90 days
    - **Business rules**: None

3.9 Autonomous AI Subsystem

3.9.1 Compute Roadmap on DAG

- **Function trigger**: Triggered automatically upon request to synthesize or restructure a learning roadmap
- **Function description**:
    - **Actors**: AI System

    - **Purpose**: Execute topological sorting over the skill DAG to generate prioritized stages and prune mastered nodes
- **Function details**:
    - **Data requirement**: skillDagGraph, initialTheta, targetScore, availableWeeks
    - **Validation**:
        - Graph must be strictly acyclic
        - Synthesized roadmap must be initialized in DRAFT state
    - **Business rules**: BR-08, BR-09, BR-14

3.9.2 Real-time Mastery & Latent Trait Estimation

- **Function trigger**: Triggered asynchronously via message queue upon every item attempt submission
- **Function description**:
    - **Actors**: AI System
    - **Purpose**: Recalculate BKT mastery probabilities and update IRT trait parameters to drive next-item selection
- **Function details**:
    - **Data requirement**: studentId, patternId, isCorrect, responseTimeSeconds
    - **Validation**:
        - Submissions with latencies < 5s have their discrimination parameter down-weighted to 0.1
        - Updated BKT states are persisted immediately to the database
    - **Business rules**: BR-06, BR-14, BR-15

3.9.3 Score Projection & Early At-Risk Detection

Chưa triển khai

4. Non-Functional Requirements

4.1 External Interfaces

- **User Interfaces:**
    - All labels, navigation items, tooltips, and system notifications shall be rendered in Vietnamese as the primary language to ensure contextual comprehension for Vietnamese high school students and parents.
    - Visual styling, including typography, responsive layouts, color schemes, and interactive elements, must strictly adhere to the project branding design guide.
    - The interface must cleanly visualize complex psychometric constructs, including Directed Acyclic Graph (DAG) roadmaps, BKT knowledge state radars, and early-warning cohort dashboards for Mentors.
- **Software Interfaces:**
    - **Artificial Intelligence Engine (LLM APIs):** Integrates with Google Gemini API via RESTful/gRPC protocols to perform document-to-item vision extraction, Socratic guidance parsing, and isomorphic item generation.
    - **Message Broker:** Integrates with RabbitMQ to handle asynchronous event streaming, including real-time BKT mastery parameter updates upon quiz submissions and background job scheduling.
    - **Notification Service:** Integrates with Firebase Cloud Messaging (FCM) and email transport services (SMTP/SendGrid) to dispatch verification OTPs,

weekly parental progress summaries, and multi-tier disciplinary escalation alerts.
    - **Payment Gateway:** Integrates with third-party payment providers (VNPay) via RESTful APIs to process online transactions for course enrollments and mock examination packages.
- **Hardware Interfaces:**
    - **Camera (Webcam / Mobile Camera):** The web and mobile applications require permission to access the client device's camera to capture periodic, random facial snapshots during mock examinations for post-hoc integrity verification.

4.2 Quality Attributes

4.2.1 Usability

- **Accessibility:** Typography, visual hierarchy, and contrast ratios must ensure high legibility and responsive layouts across modern web browsers (Chrome, Edge, Safari) and mobile operating systems (Android).
- **Error Prevention & Handling:** The system must provide instant client-side input validation and informative tooltips (e.g., password complexity, format mismatches, mandatory fields) to mitigate input mistakes before submission.
- **Learnability:** First-time users (Students, Parents, and Mentors) must be able to complete essential workflows—such as taking diagnostic tests, reviewing roadmaps, or tagging cognitive errors—without requiring exhaustive operational training manuals.
- **Operational Ergonomics for Mentors:** Mentor dashboards must employ standard traffic-light visual indicators (Green, Yellow, Red) enabling educators to identify at-risk students within 3–5 seconds of scanning.

4.2.2 Reliability

- **Data Integrity:** Assessment records, item attempt latencies, BKT latent mastery parameters, and mentor approval logs must maintain 100% ACID consistency across the database, preventing progress loss during client disconnection.
- **Availability:** The platform targets a minimum uptime of 99% during active learning hours (07:00 AM – 11:00 PM). Scheduled system maintenance and heavy data migrations must execute exclusively during off-peak hours (after 00:00 AM).
- **Recoverability:** Relational databases must run automated daily backups. In the event of system failure, the platform must guarantee a Recovery Time Objective (RTO) $\leq$ 2 hours and a Recovery Point Objective (RPO) $\leq$ 1 hour.

4.2.3 Performance

- **Response Time:**
    - Standard transactional requests (login, profile updates, individual quiz item submissions) must achieve an average latency under 2 seconds over standard 4G/Wi-Fi networks.
    - The Socratic AI Tutor must return its initial streaming response token within 3 seconds of query receipt.
    - Real-time BKT recalculation and subsequent item selection must execute within 1.5 seconds.
- **Throughput:** The backend architecture must sustain a minimum of 1,000 concurrent users taking proctored mock examinations simultaneously without service degradation or packet loss.

- **Resource Usage:** Client-side web and mobile apps must optimize memory and thread usage, ensuring webcam streaming or snapshot routines do not cause device thermal throttling or excessive battery drain.

4.2.4 Security

- **Authentication & Authorization:** All API requests must be validated using stateless JSON Web Tokens (JWT). Endpoint authorization must strictly enforce Role-Based Access Control (RBAC) across Admin, Academic Director, Academic Manager, Mentor, Student, and Parent roles.
- **Data Privacy:**
    - Passwords must be hashed with strong adaptive hashing algorithms (BCrypt/PBKDF2) using cryptographic salt before storage.
    - Parental survey insights, transcript metrics, and proctoring snapshots represent sensitive data and must be accessible strictly to authorized assigned mentors and direct legal guardians.
    - Proctoring image frames are strictly restricted to post-hoc review and must not be repurposed for automated facial recognition training.
- **Communication Security:** All traffic between clients and backend microservices must be encrypted using TLS 1.3 over HTTPS and WSS (secure WebSockets).
- **Identity Verification:** System registration and password recovery procedures require two-factor identity validation via email or SMS OTP to prevent automated bot provisioning and sybil accounts.

5. Requirement Appendix

5.1 Business Rules

ID	Category	Rule Definition
BR-01	Account & Authentication	Prospective students and parents must verify their registered email address via a One-Time Password (OTP) before initial login authorization.
BR-02	Account & Authentication	Only authenticated accounts with an `ACTIVE` status are permitted system access. Accounts with `SUSPENDED` or `BANNED` status are denied login.
BR-03	Account & Security	Passwords must contain a minimum of 8 characters, comprising at least one uppercase letter, one lowercase letter, one numeric digit, and one special character.

BR-04	Account & Security	Each email address and phone number must be unique across the platform to prevent duplicate student profiles and sybil spam accounts.
BR-05	Diagnostic Testing	The diagnostic assessment must present 30 standardized items maintaining the exact structural proportions of the official VNUHCM-EPT. It cannot be skipped during initial onboarding.
BR-06	Psychometric Modeling	The Item Response Theory (IRT 2PL) engine estimates latent competency (θ_0) along with Standard Error of Measurement (SEM). Item attempts with latency under 5 seconds ("rapid guessing") are down-weighted in trait calibration.
BR-07	Pedagogical Governance	Academic transcripts (Grades 11–12) serve as qualitative contextual reference for Mentors and shall not be ingested directly into IRT calibration formulas.
BR-08	Roadmap Lifecycle	Newly synthesized AI roadmaps are provisioned in a `DRAFT` state. A roadmap transitions to `APPROVED` only upon formal confirmation by the assigned Lead Mentor during the kickoff conference.
BR-09	Knowledge Taxonomy	The knowledge graph is structured across four canonical tiers: `Domain` → `Subject` → `Skill` → `Pattern`. All assessment items must map strictly to a single leaf `Pattern`.
BR-10	Content Verification	Item variants and external scraped questions generated by AI pipelines exist in `PRACTICE_ONLY` mode until reviewed, calibrated, and formally approved by the Academic Director.

BR-11	Content Verification	Uncalibrated AI-generated items are strictly prohibited from appearing in standardized Diagnostic Tests and Proctored Mock Examinations.
BR-12	AI Guardrails	The Socratic AI Tutor must operate under Strict Grounding, explaining steps strictly derived from verified Pattern templates without disclosing final answer keys directly.
BR-13	AI Guardrails	Final ground-truth answer keys are immutable and sourced exclusively from human-verified question banks; the AI model is prohibited from altering answer keys.
BR-14	Adaptive Mastery	The Bayesian Knowledge Tracing (BKT) engine updates latent mastery $P(L_t)$ upon each item submission. A skill node requires $P(L_t) \geq 0.85$ to achieve mastery clearance.
BR-15	Spaced Repetition	An incorrect attempt schedules isomorphic review variants of the same Pattern for delivery the following day via spaced repetition until mastery threshold criteria are met.
BR-16	Mentoring Operations	Workload constraints restrict Lead Mentors to a maximum of 15–20 assigned students. The system issues capacity threshold alerts when allocations exceed standard quotas.
BR-17	Mentoring Operations	Micro-coaching cohorts must not exceed 6 students per group, clustered by shared Pattern vulnerability vectors via K-Means clustering.
BR-18	Disciplinary Escalation (Tier 1)	Inactivity for 1–2 consecutive scheduled study days triggers automated mobile/email push notifications targeting streak preservation psychology.

BR-19	Disciplinary Escalation (Tier 2)	Inactivity reaching 3 consecutive days generates an urgent check-in task for the assigned Lead Mentor to investigate disengagement causes and adjust pacing.
BR-20	Disciplinary Escalation (Tier 3)	Inactivity reaching 5 consecutive days triggers an escalation notice to parents and Academic Managers, scheduling a mandatory three-way consultation conference.
BR-21	Exam Integrity	Mock exams enforce full-screen display modes. Exceeding 3 window/tab departures triggers an integrity infraction flag recorded in the proctoring audit log.
BR-22	Exam Integrity	Periodic webcam snapshots captured during mock exams are retained for post-hoc inspection only and must not be analyzed by automated facial AI proctoring models.
BR-23	Admissions Advisory	University major recommendations are categorized into Reach, Target, and Safety tiers evaluated strictly against score forecast intervals ($\theta \pm SEM$) rather than deterministic point thresholds.
BR-24	Remedial Interventions	A performance decline exceeding 30% or severe prerequisite concept gaps prompts the system to propose a 48-hour Remedial Sprint; activation remains subject to Mentor approval.
BR-25	Communication Governance	Exchange of external communication details or profanity in public chats is prohibited. The platform automatically redacts phone numbers, personal links, and inappropriate keywords.

5.2 Common Requirements

5.2.1. Authentication & Security

- **Unified Role-Based Access Control (RBAC):** The platform maintains a single, unified authentication portal directing users to role-specific workspaces (Admin, Academic Director, Academic Manager, Mentor, Teacher, Student, Parent) based on validated JWT claims.

- **Session Management:** Inactive user sessions are automatically invalidated after 30 minutes of idle time. Critical administrative and grading actions require re-authentication.
- **Sensitive Data Protection:** Parental surveys and adolescent study behavioral logs are encrypted at rest and accessible strictly by assigned Lead Mentors and direct legal guardians.

5.2.2. Search, Filtering & Data Exploration

- **Taxonomy-Aware Filtering:** All question banks, student cohorts, and mistake notebooks support multi-tier hierarchical filtering across Domain, Subject, Skill, and Pattern.
- **Cohort Metric Sorting:** Mentor consoles support sorting student rosters by risk level (Green/Yellow/Red), study streak, submission latency, and mastery progress percentage.
- **Pagination & Lazy Loading:** Lists containing over 20 entries (question repositories, attempt logs, proctoring snapshots) must enforce cursor-based pagination to guarantee client rendering performance.

5.2.3. Image & Document Handling

- **Examination Ingestion Validation:** Academic upload modules accept examination files in PDF and DOCX formats up to 25MB for parsing by vision-language ingestion pipelines.
- **Proctoring Snapshots:** Automated webcam capture frames are compressed to WebP format with file sizes strictly capped below 300KB per image, retaining storage efficiency across long exam sessions.

5.3 Application Messages List

#	Message Code	Message Type	Trigger Context	System Message Content
1	**MSG01**	In-page Alert	Search query or multi-tier filter yields zero results.	*Không tìm thấy kết quả phù hợp với tiêu chí đã chọn.*
2	**MSG02**	Validation Error	A mandatory input field is left empty.	*Thông tin {tên_trường} là bắt buộc.*
3	**MSG03**	Validation Error	Password length or complexity requirements are unfulfilled.	*Mật khẩu phải có ít nhất 8 ký tự, bao gồm chữ hoa, chữ thường, số và ký tự đặc biệt.*

4	**MSG04**	Toast (Success)	A roadmap is approved or student profile is initialized.	*Tạo/Thiết lập {đối_tượng} thành công.*
5	**MSG05**	Toast (Success)	Educational parameters, target scores, or taxonomies are updated.	*Cập nhật {đối_tượng} thành công.*
6	**MSG06**	Confirmation Modal	User attempts to delete a draft roadmap or purge a question pattern.	*Bạn có chắc chắn muốn xóa/hủy {đối_tượng} này? Thao tác này không thể hoàn tác.*
7	**MSG07**	Toast (Success)	Successful removal or cancellation of a task or schedule.	*Xóa/Hủy {đối_tượng} thành công.*
8	**MSG08**	Toast (Info)	Registration or password reset email dispatch.	*Mã xác thực đã được gửi đến {email/số điện thoại}. Vui lòng kiểm tra hộp thư.*
9	**MSG09**	Toast (Error)	Authentication failure due to invalid credentials.	*Email hoặc mật khẩu không chính xác. Vui lòng kiểm tra lại.*
10	**MSG10**	Warning Banner	Student attempts to switch browser tabs during a mock examination.	*Cảnh báo: Bạn vừa rời khỏi màn hình thi ({số_lần}/3). Vi phạm quá 3 lần bài thi sẽ bị hủy kết quả.*

11	**MSG11**	Toast (Warning)	Rapid response detected on an item during testing ($< 5s$).	*Bạn đang thao tác quá nhanh. Hãy đọc kỹ câu hỏi trước khi chọn đáp án.*
12	**MSG12**	Notification Modal	Level 2 or Level 3 disciplinary escalation triggered.	*Bạn đã gián đoạn chuỗi học tập {số_ngày} ngày. Một buổi hỗ trợ với Lead Mentor đã được lên lịch.*
13	**MSG13**	Toast (Success)	Student submits a report regarding an anomalous AI explanation.	*Đã ghi nhận phản ánh. Đội ngũ chuyên môn sẽ kiểm tra và hiệu chỉnh lời giải.*

IV. Software Design Description

[Provide final software design information follow the template as part II in the Report #4]

1. System Design

1.1 System Architecture

[The content of this section includes the overall diagram which includes the sub-systems, the external systems, and the relationship/connection among them. You need also provide the explanation for each of the diagram components (modules, sub-systems, external systems, etc.)].

1.2 Package Diagram

[Provide the package diagram for each sub-system. The content of this section includes overall package diagram(s) and the explanation for each package (or namespace)]

2. Database Design

[Provide the files description, database table relationship & table descriptions]

3. Detailed Design

3.1 <Feature/Function Name1>

[Provide the detailed design for the feature <Feature Name1>. It includes Class Diagram, Class Specifications, and Sequence Diagram(s); ***For the features/functions with the same structure of class & sequence diagrams, you need to provide the diagrams once for one feature/function and refer to those diagrams from other features/functions****]*

3.1.1 Class Diagram

[This part presents the class diagram for the relevant feature]

3.1.2 <Sequence Diagram Name1>

[Provide the sequence diagram(s) for the feature]

3.1.2 <Sequence Diagram Name2>

3.1.3 …

3.2 <Feature/Function Name2>

…

V. Software Testing Documentation

[Provide final software testing information follow the template as part II in the Report #5]

1. Scope of Testing

[Describe the scopes of the test. Those include the target-of-test's features, functions, and non-functional requirements that will or will not be tested.

Describe the stages/levels of testing that would be applied to your project - Unit, Integration, or System test. Each includes the in-charge, inputs/time, focuses, acceptance criteria.

List any constraints or assumptions made during the development of this document that may impact the design, development or implementation of testing]

2. Test Strategy

[List out and describe all testing types (you can refer the test types listed below or any other test types to selected the suitable ones for the project; for each selected test types you need

to provide the following information: test objective, technique, completion criteria, etc.), test levels that those test types would be performed, & the details of test supporting tools would be used in the project]

2.1 Testing Types

[List out and describe here the testing types which you would apply in your project. You need to mention following information for each type of testing: objective, technique, completion criteria]

2.2 Test Levels

<List out and describe here the testing levels which you would execute in your project. Besides, clearly state the test types which are performed in each test level that you plan for this project>

2.3 Supporting Tools

<List of the test supporting tools which will be employed for this project>

3. Test Plan

3.1 Human Resources

[List and provide the details on roles and responsibilities of the project members who would involve in testing works]

3.2 Test Environment

[List and provide the details about the tools (software, hardware, infrastructure) which the project would use for testing]

3.3 Test Milestones

[Separate test milestones, which should be identified to communicate project status accomplishments]

4. Test Cases

[Prepare the details on the test cases following the provided template

- *Unit Test Cases: Report5_Unit Test.xls*
- *Other Test Cases (IT, ST, AT): Report5_Test Report.xls]*

5. Test Reports

[Provide the test result, statistics and the relevant test analysis for your testing in the project]

VI. Release Package & User Guides

[Provide final software testing information follow the template as part II in the Report #6]

1. Deliverable Package

[The section will list all source programs, scripts, documents with version number in this release. You can see the example following table for reference, can customize or delete if not using belong to each project characteristics]

No.	Deliverable Item	Description
1	Schedule/Task Tracking	
2	Project Backlog	
3	Source Codes	
4	Database Script(s)	
5	Final Report Document	
6	Test Cases Document	
7	Defects List	
8	Issues List	
9	Slide	

2. Installation Guides

2.1 System Requirements

[Define any system requirements necessary to support the application, including the software and relevant configurations]

2.2 Installation Instruction

[Includes installation instructions and configuration guidelines]

3. User Manual

3.1 Overview

[Descript the overview of the application and if could, insert the features workflow to help user has the overview of all the features in this application]

3.2 Workflow 1

[Describe the purpose of this workflow, draw workflow diagram and other relevant diagrams]

[Describe the detailed guides for the workflow by providing the brief description, step by step guides (attached with user interface) of how to use that function]

3.3 Workflow 2

…